PetAir: A Smart Air Safety Monitoring Device for Pets

Christine Marie Lirazan
Department of Mechatronics Engineering, Kennesaw State University
clirazan@students.kennesaw.edu

***Abstract*—PetAir is an Internet of Things (IoT) device designed to improve home and pet safety by monitoring indoor conditions in real time. The system uses an ESP32 microcontroller, environmental sensors, actuators, and Arduino IoT Cloud to detect unsafe conditions and provide automatic responses. When a hazard is detected, PetAir can activate a fan, sound an alarm, change an RGB LED status indicator, and update the cloud dashboard.**

***Keywords*—Internet of Things, ESP32, Arduino IoT Cloud, Pet Safety, Smart Home**

I. INTRODUCTION

PetAir was inspired by a fire that occurred in my neighbor's townhome several months before this project began. My dog alerted my family before we realized what was happening, but the smoke and poor air quality still affected our home. Although everyone was safe, the experience showed me how quickly dangerous conditions can develop and how difficult it can be to recognize a problem when no one is home. It also made me realize how much my dog had done to protect our family, and I wanted to create something that could help keep her and other pets safe when their owners are away.

Later in the semester, I expanded this idea during a Business Fundamentals assignment, where I developed PetAir as a possible smart-home product. This project developed that concept into a working IoT prototype focused on remote monitoring and automatic safety responses.

II. METHODOLOGY

A. Hardware Design

The prototype was built using the ACEBOTT ESP32 Advanced Development Kit (26-in-1), which provided most of the required sensors and modules. The hardware includes:

- ACEBOTT ESP32 development board
- ESP32 shield board
- DHT11 temperature and humidity sensor
- Flame sensor
- DC fan with motor driver
- Active buzzer
- RGB LED
- JST cables for connecting the sensor modules

The ESP32 works as the main controller by collecting sensor data, processing system logic, controlling the connected devices, and communicating with Arduino IoT Cloud. The shield board allows the sensor modules to connect directly using JST cables, creating a cleaner and more reliable hardware setup.

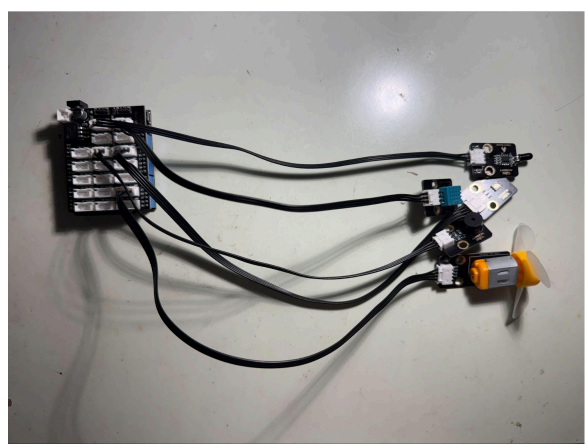

Fig. 1. Completed PetAir prototype.

B. Software and System Operation

The software was developed using the Arduino IDE and Arduino IoT Cloud libraries. The ESP32 sends sensor information to the cloud dashboard, where users can view system conditions and control selected functions.

The system operates in two modes. In automatic mode, the cooling fan activates when the temperature exceeds 75°F. In manual mode, the user can control the fan through the dashboard. If a flame is detected, the buzzer activates, the RGB LED changes to red, and the dashboard displays an unsafe condition. During normal operation, the LED displays green after showing blue during startup.

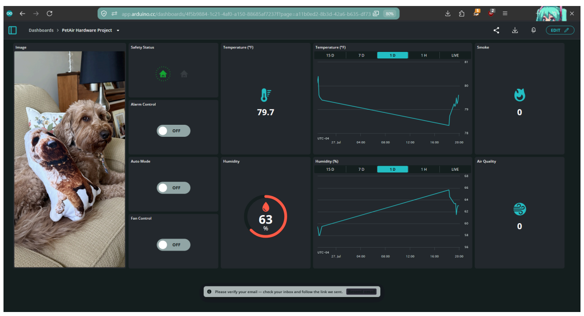

Fig. 2. Arduino IoT Cloud dashboard.

III. RESULTS AND DISCUSSION

The completed prototype successfully met the project objectives. Testing confirmed that the ESP32 could collect sensor data, communicate with Arduino IoT Cloud, and control the fan, buzzer, and RGB LED. Both automatic and manual fan control operated correctly.

Several challenges were encountered during development. The main challenge was establishing a reliable Wi-Fi connection. The ESP32 performed best when connected to a 2.4 GHz network, while higher-frequency networks were less reliable. A mobile hotspot was used during testing to monitor when the ESP32 successfully connected and synchronized with the cloud platform, which helped identify connection issues and improve the startup process. Another challenge involved time synchronization, as the ESP32 occasionally had difficulty updating the correct local time zone for recorded sensor data. This affected timestamps during testing and highlighted the importance of time synchronization in IoT systems.

Additionally, The original design included a 3D-printed enclosure; however, access to a printer was unavailable before the deadline. Instead, a cardboard enclosure was created to protect the electronics. A custom PetAir design was created in Canva, printed, and attached to improve the prototype's appearance.

A demonstration video accompanies this report and shows the prototype operating during testing.

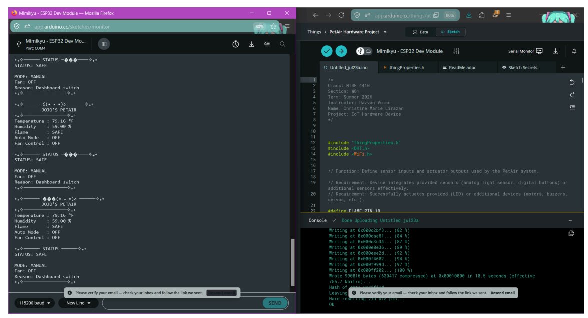

Fig. 3. PetAir operating during system testing.

Fig. 4. Final PetAir enclosure with custom Canva design.

IV. COMMERCIAL POTENTIAL

PetAir has potential as an affordable smart-home product for pet owners. According to Forbes, approximately 87 million households in the United States own at least one pet, representing a large potential market.

The completed prototype cost approximately \$49.84, including the ACEBOTT ESP32 Advanced Development Kit (\$42.85) and one month of the Arduino IoT Cloud Maker subscription (\$6.99). Assuming a retail price of \$149.00, the estimated profit per unit would be \$99.16, resulting in an estimated return on investment (ROI) of approximately 199%.

Future improvements could include a custom 3D-printed enclosure, a more customizable dashboard interface, and improved real-time data synchronization. If developed commercially, additional cloud features such as advanced analytics and expanded data storage could be offered through a subscription service to create recurring revenue.

V. CONCLUSION

In all, PetAir shows how IoT technology can help improve pet and home safety by monitoring the environment and responding to potential risks. The project successfully combined sensors, wireless communication, cloud monitoring, and device control into a working prototype. While the system achieved its main goals, future improvements could help make PetAir more reliable and practical as a smart-home safety device.

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